Distributional Treatment Effect with Latent Rank Invariance

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Distributional treatment effect

A potential outcome setup with a binary treatment: with $D_i \in \{0, 1\}$,

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

We only observe $Y_i(1)$ or $Y_i(0)$ for a given unit i.

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Existing frameworks mostly focus on some summary measures of treatment effect.

e.g.,	Average treatment effect (ATE)	$\mathbf{E}\left[Y_i(1) - Y_i(0)\right]$
	Quantile treatment effect (QTE(τ))	$F^{-1}_{Y(1)}(\tau) - F^{-1}_{Y(0)}(\tau)$

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Existing frameworks mostly focus on some summary measures of treatment effect.

Instead, I focus on **distributional treatment effect**.

e.g.,	Variance of treatment effect	$\mathrm{Var}\left(Y_i(1) - Y_i(0)\right)$
	Distribution of treatment effect	$F^{-1}_{Y(1)-Y(0)}(\tau)$

Distributional treatment effect

1. Fairness concern:

- Variance-adjusted ATE $\mathbf{E}[Y_i(1) - Y_i(0)] + \frac{\gamma}{2}\,\mathrm{Var}\left(Y_i(1) - Y_i(0)\right).$

  inequality aversion parameter $\gamma < 0$.

2. Probabilistic guarantee:

- Share of winners $1 - F_{Y(1)-Y(0)}(0).$

  When one, Pareto improving.

- α-th quantile of treatment effect $F^{-1}_{Y(1)-Y(0)}(\alpha).$

  $100 \cdot \alpha$% worst-case scenario treatment effect

3. Voluntary take-up:

- Conditional ATE given $Y_i(1) \geq Y_i(0)$ $\mathbf{E}[Y_i(1) - Y_i(0) | Y_i(1) \geq Y_i(0)].$

  policy cannot be forced at large scale.

Distributional treatment effect

Assume a **conditional independence** framework: Carneiro et al. (2003)

- A latent variable U_i such that $Y_i(1) \perp\!\!\!\perp Y_i(0) \mid U_i$. Then,

$$
\begin{aligned}
\Pr\left\{Y_i(0) \le y, Y_i(1) \le y'\right\} &= \mathbf{E}\left[\Pr\left\{Y_i(0) \le y, Y_i(1) \le y' | U_i\right\}\right] \\
&= \mathbf{E}\left[\Pr\left\{Y_i(0) \le y | U_i\right\} \cdot \Pr\left\{Y_i(1) \le y' | U_i\right\}\right].
\end{aligned}
$$

Use proxy variables X_i and Z_i to find $\Pr\{Y_i(0) \le y | U_i\}$ and $\Pr\{Y_i(1) \le y | U_i\}$.

Point identify and estimate **distributional treatment effect (DTE)** θ such that

$$
\mathbf{E}\left[m\big(Y_i(1), Y_i(0); \theta\big)\right] = 0.
$$

θ can be $\mathrm{Var}\left(Y_i(1) - Y_i(0)\right)$, $F_{Y(1)-Y(0)}(\delta)$ for a given δ and many more.

Preview of results

Identification

1. Conditional independence with two proxies: $Y_i(1), Y_i(0), X_i, \big(D_i, Z_i\big) \bigm| U_i \sim$ ind.
2. Apply diagonalization (Hu and Schennach, 2008) to untreated and treated subpopulations.
3. Connect the two decomposition results to identify $F_{Y(0),Y(1)}$.

Implementation

finite support for $U_i \Rightarrow$ conditional independence becomes finite mixture

1. First-step: nonnegative matrix factorization (NMF) for finite mixture;
   produce feasible distributions by construction.
2. Second-step: plug-in GMM for DTE, using orthogonal moments.
   first-step NMF as nuisance parameters.
   simple asymptotic standard error.

Literature review

Distributional treatment effect

- Mostly focus on partial identification, using Fréchet-Hoeffding bound or solving optimal transport

  Fan and Park (2010); Fan et al. (2014); Firpo and Ridder (2019); Frandsen and Lefgren (2021); Kaji and Cao (2023) and more.

$\Rightarrow$ Huge information gain compared to partial bounds.

- A few notable point identification exceptions:

  Heckman et al. (1997); Carneiro et al. (2003); Wu and Perloff (2006); Noh (2023).

$\Rightarrow$ DTE estimator not relying on parametric distributions nor observable conditioning variable.

Nonclassical measurement error/proximal inference/finite mixture

- Estimation often uses joint diagonalization

  Hall and Zhou (2003); Hu (2008); Hu and Schennach (2008); Kasahara and Shimotsu (2009); Bonhomme et al. (2016) and more.

$\Rightarrow$ finite mixture estimator satisfying nonnegativity constraints in a principled manner. orthogonalization procedure proposed, robust to binding nonnegativity constraints

Identification

Identification: conditional independence

An econometrican observes $\{Y_i, D_i, X_i, Z_i\}_{i=1}^n$:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0).$$

There exists a latent variable U_i.

$Y_i(1), Y_i(0), X_i, Z_i, U_i \in \mathbb{R}$ and $D_i \in \{0, 1\}$.

$\big(Y_i(1), Y_i(0), D_i, X_i, Z_i, U_i\big) \sim iid$.

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$\big(Y_i(1), Y_i(0), D_i, X_i, Z_i, U_i\big) \sim iid$.

Assumptions 1 and 2. (*conditional independence*)
$Y_i(1), Y_i(0), X_i, \big(D_i, Z_i\big)$ are mutually independent of each other given U_i.

- Only one proxy Z_i may depend on D_i given U_i.
- Can be connected to proximal inference and nonclassical measurement error literature. more
- $Y_i(1) \perp\!\!\!\perp Y_i(0) \mid U_i$ gives us key identifying power. more

Identification: what is U_i and where do we find X_i and Z_i?

1. **Measurement error model**

   In some empirical contexts, natural interpretation on U_i.

   For example, D_i is early childhood intervention and Y_i is cognitive development score.

   Then, "$U_i =$ the innate ability of a child,

   $(X_i, Z_i) =$ repeated measures of the innate ability, such as test scores"

   Carneiro et al. (2003); Cunha et al. (2010); Attanasio et al. (2020) and more.

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2. **Hidden Markov model for panel data**

   A hidden Markov model for potential outcomes:

   $$Y_{it}(d) = g_d(V_{it}, \varepsilon_{it}^d)$$

   and $\{V_{it}\}_t$ is first-order Markovian.

   Then, "$U_i =$ the contemporaneous common shock (V_{i2}),

   $(X_i, Z_i) =$ past and future outcomes (Y_{i1} and Y_{i3})."

   Kasahara and Shimotsu (2009); Hu and Shum (2012); Deaner (2023) and more

Identification: Theorem 1

Assumption 3/4. full rank/completeness of $f_{X|Z}$ when U_i is discrete/continuous: A3 A4

"Both of the proxy variables are fully informative about the latent variable U_i."

Assumption 5. $\mathbf{E}[Y_i(1)|U_i = u]$ or $\mathbf{E}[Y_i(0)|U_i = u]$ is strictly increasing in u.

"For either $d = 0$ or 1, the latent variable U_i is the rank of $\mathbf{E}[Y_i(d)|U_i]$."

U_i is a latent rank.

Theorem 1.

Assumptions 1-3 or Assumptions 1-2, 4-5 hold.

Then, the distribution of $\left(Y_i(1), Y_i(0), D_i, X_i, Z_i\right)$ is identified.

The identification extends Hu and Schennach (2008) to a potential outcome setup. more

Implementation

Implementation: finite mixture

Assume a discrete $U_i \in \{u^1, \cdots, u^K\}$ with $K < \infty$. choice of K

For continuous U_i, sieve MLE and semiparametric estimation theory:

Shen (1997); Chen and Shen (1998); Ai and Chen (2003) and more. sieve

Consider a K-way partition on the support Z_i: $\{\mathcal{Z}^j\}_{j=1}^K$.

Observable probability $\Pr\{(Y_i, X_i) \in \mathcal{Y} \times \mathcal{X} \mid D_i = d, Z_i \in \mathcal{Z}^j\}$ decomposes into

$$\begin{aligned}
&\Pr\left\{(Y_i, X_i) \in \mathcal{Y} \times \mathcal{X} \,\middle|\, D_i = d, Z_i \in \mathcal{Z}^j\right\} \\
&= \sum_{k=1}^{K} \underbrace{\Pr\left\{Y_i(d) \in \mathcal{Y} \,\middle|\, U_i = u^k\right\} \cdot \Pr\left\{X_i \in \mathcal{X} \,\middle|\, U_i = u^k\right\}}_{=\text{mixture component probability}} \cdot \underbrace{\Pr\left\{U_i = u^k \,\middle|\, D_i = d, Z_i \in \mathcal{Z}^j\right\}}_{=\text{mixture weights}}
\end{aligned}$$

for $j = 1, \ldots, K$.

Implementation: finite mixture

By stacking up mixture weights across the partition, for each treatment status,

$$\Lambda_d = \begin{pmatrix} \Pr\left\{U_i = u^1 \middle| D_i = d, Z_i \in \mathcal{Z}^1\right\} & \cdots & \Pr\left\{U_i = u^1 \middle| D_i = d, Z_i \in \mathcal{Z}^K\right\} \\ \vdots & \ddots & \vdots \\ \Pr\left\{U_i = u^K \middle| D_i = d, Z_i \in \mathcal{Z}^1\right\} & \cdots & \Pr\left\{U_i = u^K \middle| D_i = d, Z_i \in \mathcal{Z}^K\right\} \end{pmatrix}.$$

Then, for any $\mathcal{Y}$ and $d = 0, 1$,

$$\begin{pmatrix} \Pr\left\{Y_i \in \mathcal{Y} \middle| D_i = d, Z_i \in \mathcal{Z}^1\right\} \\ \vdots \\ \Pr\left\{Y_i \in \mathcal{Y} \middle| D_i = d, Z_i \in \mathcal{Z}^K\right\} \end{pmatrix}^{\intercal} = \begin{pmatrix} \Pr\left\{Y_i(d) \in \mathcal{Y} \middle| U_i = u^1\right\} \\ \vdots \\ \Pr\left\{Y_i(d) \in \mathcal{Y} \middle| U_i = u^K\right\} \end{pmatrix}^{\intercal} \Lambda_d.$$

Assumption 3 assume that Λ_0 and Λ_1 are full rank.

The distribution of $Y_i(d)$ given U_i is a function of observable distributions and $(\Lambda_d)^{-1}$.

Implementation: finite mixture

Proposition 1.

Assumptions 1-3 hold. A parameter of interest θ is identified by an infeasible moment condition

$$\mathbf{E}\left[m\big(Y_i(1), Y_i(0), D_i, X_i; \theta\big)\right] = 0.$$

Then, θ is identified by a feasible quadratic moment function $\tilde{m}$:

$$\mathbf{E}\left[\tilde{m}\left(\big(Y_i, D_i, X_i, Z_i\big), \big(Y_{i'}, D_{i'}, X_{i'}, Z_{i'}\big); \theta, \Lambda_0, \Lambda_1\right)\right] = 0$$

where $i \neq i'$. example

The proxy Z_i is used to shift U_i.

We need two observations since m depends on both $Y_i(1)$ and $Y_i(0)$.

This builds on Bonhomme et al. (2017).

Implementation

To do a plug-in estimation, I orthogonalize $\tilde{m}$:

$$\mathbf{E}\left[\psi\big((Y_i, D_i, X_i, Z_i), (Y_{i'}, D_{i'}, X_{i'}, Z_{i'}); \theta, \Lambda_0, \Lambda_1\big)\right] = 0.$$

Estimation error in $\big(\Lambda_0, \Lambda_1\big)$ has no first-order impact. more

A feasible GMM model for DTE parameter θ with

$$\mathbf{E}\left[\psi\big((Y_i, D_i, X_i, Z_i), (Y_{i'}, D_{i'}, X_{i'}, Z_{i'}); \theta, \Lambda_0, \Lambda_1\big)\right] = 0.$$

Step 1. Estimate the nuisance parameter $\Lambda_d = \Big(f_{U|D,Z}(u|d,z)\Big)_{u,z}$.

Step 2. Plug-in GMM to estimate θ.

Implementation, step 1: nonnegative matrix factorization

To estimate the finite mixture weights, construct additional partitions: with $M_Y \geq 2$ and $M_X \geq K$,

$$\{\mathcal{Y}^m\}_{m=1}^{M_Y}, \quad \{\mathcal{X}^m\}_{m=1}^{M_X}, \quad \text{and} \quad \{\mathcal{Z}^j\}_{j=1}^{K}.$$

Let $\Gamma_{Y(0)} = \left(\Pr\left\{ Y_i(0) \in \mathcal{Y}^m \middle| U_i = u^k \right\} \right)_{m,k}$ and similarly define $\Gamma_{Y(1)}$ and Γ_X.

Estimate Λ_0, Λ_1 by solving

$$\min \sum_{m,d,m',j} \left(\frac{\sum_{i=1}^n \mathbf{1}\{Y_i \in \mathcal{Y}^m, D_i = d, X_i \in \mathcal{X}^{m'}, Z_i \in \mathcal{Z}^j\}}{\sum_{i=1}^n \mathbf{1}\{D_i = d, Z_i \in \mathcal{Z}^j\}} - \sum_{k=1}^{K} [\Gamma_{Y(d)}]_{m,k} \cdot [\Gamma_X]_{m',k} \cdot [\Lambda_d]_{k,j} \right)^2$$

subject to 1) $\Gamma_{Y(0)}, \Gamma_{Y(1)}, \Gamma_X, \Lambda_0, \Lambda_1$ are nonnegative and
2) their columnwise sums are one.

This factorizes $\left(\Pr\left\{ Y_i \in \mathcal{Y}^m, X_i \in \mathcal{X}^{m'} \,\middle|\, D_i = d, Z_i \in \mathcal{Z}^j \right\} \right)_{m,m',d,j}$ into nonnegative matrices.

Implementation, step 1: nonnegative matrix factorization

Nonparametric estimation of finite mixture.

The optimization is tri-convex; estimation is done iteratively, only attaining local minima. algorithm

We may use joint diagonalization or unconstrained NMF to initialize.

At the cost of higher computational burden, validity in finite sample without ex-post refinement.

1. The mixture weights are nonnegative.
2. Retrieved conditional density of $Y_i(d)$ given U_i is nonnegative when integrated over $\{\mathcal{Y}^m\}_{m=1}^{M_Y}$.

The ingredients (e.g., $f_{Y(1)|U}, f_{Y(0)|U}$) of DTE parameter θ is a valid density function.

For example, for the share of losers

$$F_{Y(1)-Y(0)}(0) = \sum_{k=1}^{K} \Pr\{U_i = u^k\} \int_{\mathbb{R}} f_{Y(0)|U}(y|u^k) \cdot F_{Y(1)|U}(y|u^k) dy.$$

negative density estimate of $f_{Y(0)|U}$ on the right tail may distort the DTE by much.

Implementation, step 2: plug-in GMM

Regularity conditions for the infeasible GMM model are assumed. more

Theorem 3.
Assumptions 1-3 hold. $\hat{\theta}$ is the plug-in GMM estimator of θ, using ψ. Then,

$$\sqrt{n}\left(\hat{\theta}-\theta\right) \xrightarrow{d} \mathcal{N}\left(0,\Sigma\right)$$

as $n \to \infty$ with some consistently estimable Σ.

Inference is computationally easy thanks to orthogonalization;
consistent estimator for the asymptotic standard error.

For any moment-identified DTE parameter θ, the implementation is readily available.

Implementation: falsification test

Linear constraint $X_i \perp\!\!\!\perp D_i \mid U_i$ from Assumption 1 connects $\{i : D_i = 1\}$ and $\{i : D_i = 0\}$.
NMF is still well-posed without the linear constraint.

Thus, we can 1) introduce two separate Γ_X for each treatment status in NMF and;
2) test $X_i \perp\!\!\!\perp D_i \,\big|\, U_i$ as a falsification test:

$$\sum_{k=1}^{K-1} \sum_{m=1}^{M_X-1} \left(f_{X|D=1,U}\left(x^m | u^k\right) - f_{X|D=0,U}\left(x^m | u^k\right) \right)^2 = 0.$$

Theorem 4.
Under Assumptions 1-3, $T_n \xrightarrow{d} \chi^2\big((K-1)\cdot(M_X-1)\big)$ as $n \to \infty$.

Simulation

Simulation

Monte Carlo simulations ($B = 2000$) with three DGPs such that $X_i, Z_i, U_i \in \{1, 2, 3\}$.

$Y_i(d) \mid (U_i = k) \sim \mathcal{N}\left(\mu^k(d), \sigma^k(d)^2\right)$ with

$$\begin{pmatrix} \mu^1(0) & \sigma^1(0) \\ \mu^2(0) & \sigma^2(0) \\ \mu^3(0) & \sigma^3(0) \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \mu^1(1) & \sigma^1(1) \\ \mu^2(1) & \sigma^2(1) \\ \mu^3(1) & \sigma^3(1) \end{pmatrix} = \begin{pmatrix} 1.5 & 1.5 \\ 2 & 1 \\ 2.5 & 0.5 \end{pmatrix}.$$

$\Lambda = \Big(\Pr\{Z_i = z \,|\, U_i = k\} \Big)_{z,k}$ is

$$\begin{pmatrix} 0.516 & 0.237 & 0.161 \\ 0.323 & 0.526 & 0.323 \\ 0.161 & 0.237 & 0.516 \end{pmatrix}, \quad \begin{pmatrix} 0.630 & 0.181 & 0.120 \\ 0.250 & 0.638 & 0.250 \\ 0.120 & 0.181 & 0.630 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0.747 & 0.127 & 0.077 \\ 0.176 & 0.746 & 0.176 \\ 0.077 & 0.127 & 0.747 \end{pmatrix}.$$

The smallest singular values for Λ are 0.202, 0.386 and 0.568. specifics

DTE parameter is marginal distribution of treatment effect $F_{Y(1)-Y(0)}(\delta)$.

Simulation

	true value	bias					
$\widehat{F}_{Y(1)-Y(0)}(0)$	0.084	-0.010	-0.001	-0.001	0.003	0.001	0.000
$\widehat{F}_{Y(1)-Y(0)}(1)$	0.264	-0.004	0.001	-0.000	0.003	0.001	0.000
$\widehat{F}_{Y(1)-Y(0)}(2)$	0.536	0.004	-0.001	-0.002	0.001	-0.001	-0.000
$\widehat{F}_{Y(1)-Y(0)}(3)$	0.775	0.016	0.002	0.001	0.002	-0.001	-0.000
$\widehat{F}_{Y(1)-Y(0)}(4)$	0.911	0.028	0.009	0.005	0.006	0.001	0.001
$\sigma_{\min}(\Lambda)$		0.202	0.386	0.568	0.202	0.386	0.568
n		500	500	500	2000	2000	2000

Table 1: Bias of DTE estimator $\widehat{F}_{Y(1)-Y(0)}(\delta)$.

Estimates are projected onto $[0, 1]$.

Estimation performance improves as Z_i gets more informative, i.e. $\sigma_{\min}(\Lambda)$ goes up.

Simulation

	true value	rMSE					
$\widehat{F}_{Y(1)-Y(0)}(0)$	0.084	0.044	0.022	0.018	0.015	0.011	0.009
$\widehat{F}_{Y(1)-Y(0)}(1)$	0.264	0.078	0.032	0.028	0.022	0.017	0.015
$\widehat{F}_{Y(1)-Y(0)}(2)$	0.536	0.085	0.037	0.033	0.023	0.019	0.017
$\widehat{F}_{Y(1)-Y(0)}(3)$	0.775	0.080	0.031	0.027	0.019	0.015	0.013
$\widehat{F}_{Y(1)-Y(0)}(4)$	0.911	0.062	0.023	0.018	0.016	0.010	0.008
$\sigma_{\min}(\Lambda)$		0.202	0.386	0.568	0.202	0.386	0.568
n		500	500	500	2000	2000	2000

Table 2: RMSE of DTE estimator $\widehat{F}_{Y(1)-Y(0)}(\delta)$.

Estimates are projected onto $[0,1]$.

Estimation performance improves as Z_i gets more informative, i.e. $\sigma_{\min}(\Lambda)$ goes up.

Simulation

	true value	coverage probability					
$\widehat{F}_{Y(1)-Y(0)}(0)$	0.084	0.988	0.986	0.956	0.992	0.965	0.947
$\widehat{F}_{Y(1)-Y(0)}(1)$	0.264	0.992	0.986	0.969	0.992	0.969	0.954
$\widehat{F}_{Y(1)-Y(0)}(2)$	0.536	0.992	0.985	0.967	0.989	0.964	0.951
$\widehat{F}_{Y(1)-Y(0)}(3)$	0.775	0.992	0.980	0.956	0.990	0.962	0.946
$\widehat{F}_{Y(1)-Y(0)}(4)$	0.911	0.980	0.966	0.946	0.980	0.954	0.943
$\sigma_{\min}(\Lambda)$		0.202	0.386	0.568	0.202	0.386	0.568
n		500	500	500	2000	2000	2000

Table 3: Coverage of 95% confidence interval.

Conservatism when $\sigma_{\min}(\Lambda)$ is low and n is small.

Estimation error in $\widehat{\Lambda}$ is further amplified through inversion when $\sigma_{\min}(\Lambda)$ is low.

Empirical Illustration

Empirical illustration: setup

I revisit Jones et al. (2019), which studies the effect of workplace wellness program.
The program *eligibility* was randomly assigned to employees at UIUC; intent-to-treat.

The variables in the dataset are:

$$Y_i = \text{monthly medical spending over August 2016-July 2017}$$

$$D_i = \mathbf{1}\{\text{eligible for the wellness program starting in September 2016}\}$$

$$X_i = \text{monthly medical spending over July 2015-July 2016}$$

$$Z_i = \text{monthly medical spending over August 2017-January 2019}$$

"This year's underlying health status U_i only depends on last year's health status."

Empirical illustration: treatment effect distribution

Rank test suggests $K \geq 3$. choice of K

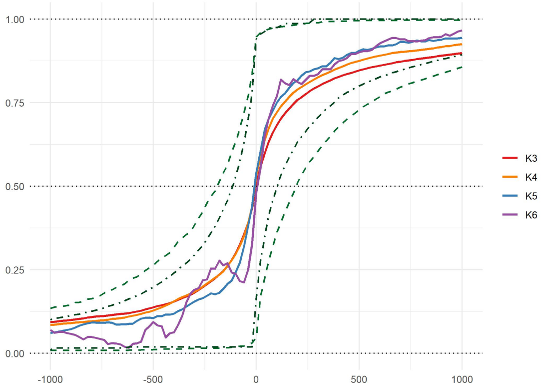


Figure 1: Marginal distribution of $Y_i(1) - Y_i(0)$, across K.

For 37% of the support, $\widehat{F}_{Y(1)-Y(0)}$ with $K = 6$ was decreasing.

Possible misspecification/discretization bias on the right tail.

Empirical illustration: treatment effect distribution

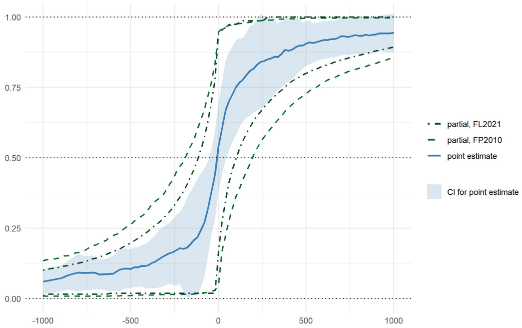


Figure 2: Marginal distribution of $Y_i(1) - Y_i(0)$.

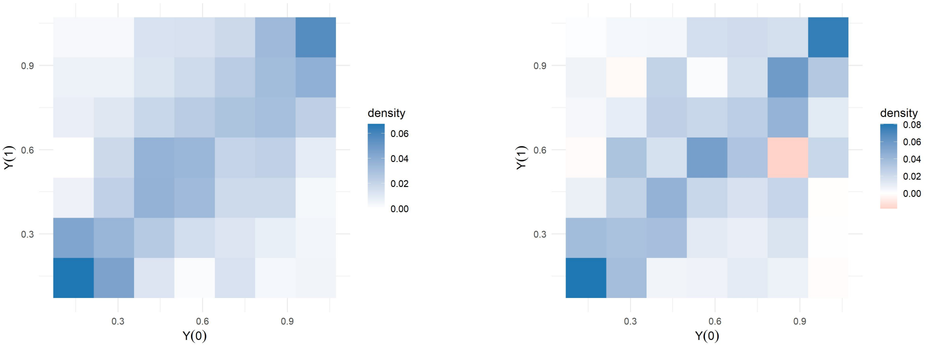
Fan and Park (2010): $\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i$.

Frandsen and Lefgren (2021): $\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i$,

$F_{Y(1)|Y(0)}(y|y')$ is decreasing in y' for all y,

$F_{Y(0)|Y(1)}(y|y')$ is decreasing in y' for all y.

Empirical illustration: joint density of $Y_i(1)$ and $Y_i(0)$



(a) $K = 4$ (b) $K = 5$

Figure 3: Joint density of $Y_i(1)$ and $Y_i(0)$, across $K = 4, 5$.

High correlation on the two ends of the spectrum.

Conclusion

- Assume a latent variable U such that

$$Y_i(1) \perp\!\!\!\perp Y_i(0) \,\Big|\, U_i.$$

  This assumption could be thought of as a 'latent rank invariance' condition.

- Use two proxy variables X_i and Z_i to identify the distribution of $Y_i(d)|U_i$.
  Measurement error model and hidden Markov model motivate the use of proxies.

- Nonnegative matrix factorization estimates finite mixture.
  Better finite sample performance due to additional regularization.

- An asymptotic distribution is derived for the DTE estimator.

Focus on distributional treatment effect

Summary measures are invariant the dependence structure between $Y_i(1)$ and $Y_i(0)$.

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1. Fairness concern: Levy and Markowitz (1979); Epstein and Segal (1992) and more

   "Aggregate treatment effect with inequality aversion."

   "I'd like to look at treatment effect heterogeneity w.r.t. baseline."

   - Variance-adjusted ATE $\mathbf{E}[Y_i(1) - Y_i(0)] + \frac{\gamma}{2}\,\mathrm{Var}\left(Y_i(1) - Y_i(0)\right)$.
   - Conditional ATE given $Y_i(0) \in \mathcal{Y}$ $\mathbf{E}[Y_i(1) - Y_i(0) | Y_i(0) \in \mathcal{Y}]$.

Focus on distributional treatment effect

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   "Aggregate treatment effect with inequality aversion."

   "I'd like to look at treatment effect heterogeneity w.r.t. baseline."

2. Probabilistic guarantee: Reeve et al. (2023); Chernozhukov et al. (2025)

   "How many people are better off under treatment?"

   "What is $100 \cdot \alpha$% worst-case scenario?"

   - Share of winners $1 - F_{Y(1)-Y(0)}(0)$.
   - α-th quantile of treatment effect $F^{-1}_{Y(1)-Y(0)}(\alpha)$.

Focus on distributional treatment effect

Summary measures are invariant the dependence structure between $Y_i(1)$ and $Y_i(0)$.

Can't answer questions on treatment effect distribution or joint distribution of $Y_i(1)$ and $Y_i(0)$:

1. **Fairness concern:** Levy and Markowitz (1979); Epstein and Segal (1992) and more

   "Aggregate treatment effect with inequality aversion."

   "I'd like to look at treatment effect heterogeneity w.r.t. baseline."

2. **Probabilistic guarantee:** Reeve et al. (2023); Chernozhukov et al. (2025)

   "How many people are better off under treatment?"

   "What is $100 \cdot \alpha$% worst-case scenario?"

3. **Voluntary take-up:** Heckman and Vytlacil (2005); Mogstad et al. (2018) and more

   "How many people would opt into treatment at the cost of c."

   "Policy-relevant treatment effect when people can opt out with full information."

   - Take-up at cost c $\quad F_{Y(1)-Y(0)}(c)$.
   - Conditional ATE given $Y_i(1) \geq Y_i(0)$ $\quad \mathbf{E}[Y_i(1) - Y_i(0) | Y_i(1) \geq Y_i(0)]$.

back

Conditional independence framework

Assumption 1. $(Y_i(1), Y_i(0), X_i) \perp\!\!\!\perp (D_i, Z_i) \mid U_i$.

- In proximal inference literature,
  $X_i =$ outcome-aligned proxy and $Z_i =$ treatment-aligned proxy.
  Miao et al. (2018); Deaner (2023); Nagasawa (2022) and more.

- Distribution of $(Y_i(d), X_i) \mid U_i$ is set identified: Henry et al. (2014).

Assumption 2. $Y_i(1), Y_i(0), X_i$ are mutually independent given U_i.

- Nonclassical measurement error literature: $Y_i, X_i, Z_i \mid U_i \sim$ ind.
  Hu (2008); Hu and Schennach (2008) and more.

- Conditional distributions are point identified.

- Extended to a potential outcome setup.
  Additionally, $Y_i(1) \perp\!\!\!\perp Y_i(0) \mid U_i$ recovers the joint dist. of $Y_i(1)$ and $Y_i(0)$.

back

Conditional independence: example 1 *(repeated measurement)*

Attanasio et al. (2020): early childhood intervention's effect on children's development.

Y_i is test score at follow-up, U_i is innate ability at baseline, and $\left(X_i, Z_i\right)$ are test scores at baseline.

$$
\begin{aligned}
Y_i(d) &= \mu^d + \alpha^d U_i + \varepsilon_i^d \quad \text{for } d = 0, 1, \\
X_i &= \mu^X + \alpha^X U_i + \varepsilon_i^X, \\
Z_i &= \mu^Z + \alpha^Z U_i + \varepsilon_i^Z.
\end{aligned}
$$

Conditional independence: example 1 *(repeated measurement)*

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Assumptions 1-2 hold when

- $\varepsilon_i^0, \varepsilon_i^1, \varepsilon_i^X$ and ε_i^Z are mutually independent given U_i.
- D_i is randomly assigned.

back

Conditional independence: example 2 *(past and future outcomes)*

A common shock process $\{V_{it}\}_{t=1}^{3}$ and random shocks $\left(\varepsilon_{i1}^{0}, \varepsilon_{i2}^{0}, \varepsilon_{i2}^{1}, \varepsilon_{i3}^{0}, \varepsilon_{i3}^{1}\right)$.

$$Y_{it}(d) = g_d\left(V_{it}, \varepsilon_{it}^{d}\right) \quad \text{for } d = 0, 1 \text{ and } t = 1, 2, 3,$$

$$Y_{it} = \begin{cases} Y_{i1}(0) & \text{if } t = 1 \\ D_i \cdot Y_{it}(1) + (1 - D_i) \cdot Y_{it}(0) & \text{if } t \geq 2 \end{cases}.$$

Conditional independence: example 2 *(past and future outcomes)*

A common shock process $\{V_{it}\}_{t=1}^{3}$ and random shocks $\left(\varepsilon_{i1}^{0}, \varepsilon_{i2}^{0}, \varepsilon_{i2}^{1}, \varepsilon_{i3}^{0}, \varepsilon_{i3}^{1}\right)$.

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$$Y_{it} = \begin{cases} Y_{i1}(0) & \text{if } t = 1 \\ D_i \cdot Y_{it}(1) + (1 - D_i) \cdot Y_{it}(0) & \text{if } t \geq 2 \end{cases}.$$

Assumptions 1-2 hold with $\left(Y_i, X_i, Z_i, U_i\right) = \left(Y_{i2}, Y_{i1}, Y_{i3}, V_{i2}\right)$ when

- $\left(\{V_{it}\}_{t=1}^{3}, D_i\right), \varepsilon_{i1}^{0}, \varepsilon_{i2}^{0}, \varepsilon_{i2}^{1}, \varepsilon_{i3}^{0}, \varepsilon_{i3}^{1}$ are mutually independent.
- $\{V_{it}\}_{t=1}^{3}$ is first-order Markovian given D_i: $\quad V_{i1} \perp\!\!\!\perp V_{i3} \mid (V_{i2}, D_i)$.
- D_i is randomly assigned at time $t = 2$: $\{V_{it}\}_{t=1}^{2} \perp\!\!\!\perp D_i$.

back

Regime-changing treatment effect mechanism

In both measurement error model and hidden Markov model,

- Two separate outcome generating processes for $Y_i(1)$ and $Y_i(0)$: *regime-changing*.

$$Y_i(0) = g_0(U_i, \varepsilon_i^0),$$
$$Y_i(1) = g_1(U_i, \varepsilon_i^1).$$

The regime-specific random shocks are purely random, satistfying $\varepsilon_i^1 \perp\!\!\!\perp \varepsilon_i^0 \mid U_i$.
In contrast to *input-changing* treatment mechanism. more

back

Examples of regime-changing treatment mechanism

Thus, Assumption 2 is most plausible when the treatment induces systemic changes:

- Attanasio et al. (2020):
  Treatment provided parenting guidance, changing how parents interacted with children.
- Jones et al. (2019): $\leftarrow$ my empirical example
  Treatment provided information sessions on healthy lifestyle,
  changing how participants sought medical service and took self-care measures.
- Job assignment: e.g. the National Supported Work Demonstration.
  Treatment changes how worker skill U_i leads to outcome Y_i such as income.
- Teaching methodology: e.g. Banerjee et al. (2007); Muralidharan et al. (2019).
  Treatment changes how student aptitude U_i leads to outcome Y_i such as academic achievement.

back

Input-changing treatment mechanism

Two common independence assumptions:

$$Y_i(1) \perp\!\!\!\perp Y_i(0) \quad \text{and} \quad Y_i(0) \perp\!\!\!\perp \big(Y_i(1) - Y_i(0)\big)$$

The latter can be motivated by *input-changing* treatment mechanism: with $V_i \perp\!\!\!\perp \varepsilon_i \mid U_i$,

$$Y_i(d) = \alpha + \mu^0 U_i + d \cdot \mu^1 V_i + \varepsilon_i$$

satisfies $Y_i(0) \perp\!\!\!\perp \big(Y_i(1) - Y_i(0)\big) \mid U_i$.

Treatment turns on a new source of individual-level heterogeneity V_i,
which is (conditionally) independent of the existing heterogeneity ε_i.

For example, providing new infrastructure: computers in teaching environment. back

Spectral Theorem of Hu and Schennach (2008)

A linear operator $L_{Y=y,X|D=d,X}$ maps a density of Z_i to a density of $(Y_i(d)=y, X_i)$:

$$\left(L_{Y=y,X|D=d,Z}\, g\right)(x) = \int_{\mathbb{R}} f_{Y(d),X|D,Z}(y,x|d,z)g(z)dz.$$

From the decomposition based on Assumption 2, we get

$$L_{Y=y,X|D=d,Z} = L_{X|U} \cdot \Delta_{Y=y|U} \cdot L_{U|D=d,Z}$$

with similarly defined operators $L_{X|U}$, $L_{U|D=d,Z}$ and a diagonal operator $\Delta_{Y=y|U}$. Thus,

$$\begin{aligned} L_{Y=y,X|D=d,Z}\left(L_{X|D=d,Z}\right)^{-1} &= L_{X|U} \cdot \Delta_{Y=y|U} \cdot L_{U|D=d,Z} \cdot \left(L_{X|U} \cdot L_{U|D=d,Z}\right)^{-1} \\ &= \underbrace{L_{X|U} \cdot \Delta_{Y=y|U} \cdot \left(L_{X|U}\right)^{-1}}_{\text{spectral decomposition}}. \end{aligned}$$

back

Assumptions 1-3

Assumptions 1-2.
$Y_i(1), Y_i(0), X_i, (D_i, Z_i)$ are mutually independent of each other given U_i.

Assumption 3.

a. *(finitely discrete U_i)* $U_i \in \{u^1, \cdots, u^K\}$.

b. *(full rank)* Γ_X, Λ_0 and Λ_1 have rank K.

c. *(no repeated eigenvalue)* For any $k \neq k'$, there exist some $d \in \{0, 1\}$ and y such that

$$\Pr\left\{Y_i(d) = y \middle| U_i = u^k\right\} \neq \Pr\left\{Y_i(d) = y \middle| U_i = u^{k'}\right\}.$$

A3 P1 T2

Assumption 4.

a. *(continuous U_i)* $U_i \in [0,1]$.

b. *(bounded density)* All marginal and conditional densities of $\big(Y_i(1), Y_i(0), X_i, Z_i, U_i\big)$ are bounded.

c. *(completeness)* Let $f_{X|Z,d}$ denote the conditional density of X_i given $(D_i = d, Z_i)$.

$$\int_{\mathbb{R}} |g(x)|dx \quad \text{and} \quad \int_{\mathbb{R}} g(x) f_{X|Z,d}(x|z)d(x) = 0 \quad \forall d, z$$

implies $g(x) = 0$. Assume similarly for $f_{X|U}$.

d. *(no repeated eigenvalue)* $\forall u \neq u'$, there exists $d \in \{0,1\}$ such that

$$\Pr\left\{f_{Y(d)|U}(Y_i(d)|u) \neq f_{Y(d)|U}(Y_i(d)|u')|D_i = d\right\} > 0.$$

Why do we need Assumption 5?

Under Assumptions 1,2 and 3/4, we have identified

$$\{f_{Y(1)|U}(\cdot|u), f_{Y(0)|U)}(\cdot|u), f_{X|U}(\cdot|u), f_{U|D=1,Z}(u|\cdot), f_{U|D=0,Z}(u|\cdot)\}_u.$$

Each group of five is not yet connected to a value of u; unordered collection.

When the collection is finite, not having an ordering is okay.
Probability mass function of U_i is still identified.

When the collection is infinite, not having an ordering matters.
Why? Now we are in density territory.
$\{f_{U|D=1,Z}(u|\cdot), f_{U|D=0,Z}(u|\cdot)\}_u$ gives us $\{f_U(u)\}_u$ but not f_U.
Assumption 5 orders $\{f_U(u)\}_u$ using $\tilde{u} = \mathbf{E}[Y_i(d)|U_i = u]$. back

Diagonalization à la Hu and Schennach (2008)

Given (y, d), construct

$$\mathbf{H}_d(y) = \begin{pmatrix} \Pr\left\{Y_i = y, X_i = x^1 \middle| D_i = d, Z_i = z^1\right\} & \cdots & \Pr\left\{Y_i = y, X_i = x^1 \middle| D_i = d, Z_i = z^J\right\} \\ \vdots & \ddots & \vdots \\ \Pr\left\{Y_i = y, X_i = x^{M_X} \middle| D_i = d, Z_i = z^1\right\} & \cdots & \Pr\left\{Y_i = y, X_i = x^{M_X} \middle| D_i = d, Z_i = z^J\right\} \end{pmatrix}.$$

Y_i, X_i, Z_i can be discretized when continuous.

Suppose U_i is discrete. Under Assumptions 1-2 that $Y_i(1), Y_i(0), X_i, (D_i, Z_i) \mid U_i \sim$ ind

$$\mathbf{H}_d(y) = \Gamma_X \cdot \Delta_d(y) \cdot \Lambda_d$$

where $\Gamma_X = \left(\Pr\left\{X_i = x^m \middle| U_i = u^k\right\}\right)_{m,k}$

$\Delta_d(y) = \text{diag}\left(\Pr\left\{Y_i(d) = y \middle| U_i = u^k\right\}\right)_k$

$\Lambda_d = \left(\Pr\left\{U_i = u^k \middle| D_i = d, Z_i = z^j\right\}\right)_{k,j}$. back

Diagonalization à la Hu and Schennach (2008)

$\sum_{y'} \mathbf{H}_d(y') = \Gamma_X \cdot \left(\sum_{y'} \Delta_d(y') \right) \cdot \Lambda_d = \Gamma_X \cdot \Lambda_d.$

When $\Gamma_X \cdot \Lambda_d$ is invertible, we get

$$\mathbf{H}_d(y) \Big(\sum_{y'} \mathbf{H}_d(y') \Big)^{-1} = \Gamma_X \cdot \Delta_d(y) \cdot \Lambda_d \cdot \Big(\Gamma_X \cdot \Lambda_d \Big)^{-1} = \Gamma_X \cdot \Delta_d(y) \cdot \Big(\Gamma_X \Big)^{-1}$$

Eigenvalue decomposition finds $\Delta_d(y)$ and Γ_X up to sign and scale.
Repeating this across y completes identification: Hu (2008)

Hu and Schennach (2008) develops its counterpart for continuous U_i. more
Conditional densities $f_{X|U}, f_{Y(d)|U}$ are identified.

d is fixed; needs to extend Hu and Schennach (2008) to potential outcome setup. back

Sketchy of proof

1. Apply Hu and Schennach (2008) to the two subpopulations:

$$(\Gamma_X, \{\Delta_0(y)\}_y) \quad \text{and} \quad (\Gamma_X, \{\Delta_1(y)\}_y)\,.$$

   Labelings on U_i are connected using Γ_X since $X_i \perp\!\!\!\perp D_i \mid U_i$.

2. $\{\Delta_0(y)\}_y$ give us $f_{Y(0)|U}$ and $\{\Delta_1(y)\}_y$ give us $f_{Y(1)|U}$.

3. Given Γ_X, we get

$$\Lambda_d = \left(f_{U|D,Z}(u|d,z)\right)_{u,z} = (\Gamma_X)^+ \sum_{y'} \mathbf{H}_d(y').$$

   Λ_0, Λ_1 and the observed distribution of (D_i, Z_i) identifies the distribution of U_i:

$$f_U(u) = \mathbf{E}\left[f_{U|D,Z}(u|D_i, Z_i)\right].$$

$\Rightarrow f_{Y(1)|U}, f_{Y(0)|U}, f_U$ are identified. $Y_i(1) \perp\!\!\!\perp Y_i(0) \mid U_i$ identifies $F_{Y(0),Y(1)}$. back

Multidimensional U_i

Skill formation/human capital accumulation literature often model two-dimensional U_i:

Carneiro et al. (2003); Cunha et al. (2010); Attanasio et al. (2020) and more

- $U_i = (U_i^C, U_i^N)$: cognitive and noncognitive ability of a child.
- $X_i = (X_i^C, X_i^N)$: cognitive ability test scores and noncognitive ability test scores.
- $Z_i = (Z_i^C, Z_i^N)$: another set of ability scores, measured independently.

Components of X_i, Z_i need not match components of U_i.

Helps in assuming that any remaining heterogeneity after controlling for U_i is purely random.

back

Identification: implicit restriction

A crucial step in the identification argument is that there exists some w such that

$$\mathbf{E}[Y_i(1)|Y_i(0)=y] = \int_{\mathbb{R}} \frac{w(y,z)}{f_{Y(0)}(y)} \cdot \mathbf{E}[Y_i|D_i=1,Z_i=z]dz,$$

$$\mathbf{E}[Y_i(1)Y_i(0)] = \int_{\mathbb{R}}\int_{\mathbb{R}} w(y,z)\cdot y\mathbf{E}[Y_i|D_i=1,Z_i=z]dydz.$$

$\mathbf{E}[Y_i|D_i=1,Z_i]$ replaces $Y_i(1)$ and $w(y,z)$ replaces the joint density of $\big(Y_i(1),Y_i(0)\big)$.
"Proxy variable Z_i creates sufficient variation in the distribution of $Y_i(1)$."

The implicit restriction is that
"conditional distribution of $Y_i(1)$ given $Y_i(0)$ is a linear combination of $\{F_{Y|D=1,Z}(\cdot|z)\}_z$."

Sieve MLE

To allow for a continuous U_i, we can directly construct a likelihood using sieves:

$$f_{Y,X|D=d,Z,n}(y,x|z;\theta) = \int_{\mathbb{R}} f_{Y(d)|U,n}(y|u;\theta) \cdot f_{X|U,n}(x|u;\theta) \cdot f_{U|D=d,Z,n}(u|z;\theta)du.$$

Nonnegativity, sum-to-one, monotonicity conditions are easy to impose with Bernstein polynomials: a Bernstein polynomial of degree m is

$$g_m(u) = \sum_{k=0}^{m} \theta_k u^k (1-u)^{m-k}.$$

Then, monotonicity of $\int_0^1 u g_m(u)du$ is a set of linear constraints on $\{\theta_k\}_{k=0}^m$.

back

Choice of K

Under Assumption 3, the rank of the following $M_X \times 2M_Z$ matrix is K:

$$\mathbf{H}_X = \begin{pmatrix} \Pr\left\{X_i \in \mathcal{X}^1 \middle| D_i = 0, Z_i \in \mathcal{Z}^1\right\} & \cdots & \Pr\left\{X_i \in \mathcal{X}^1 \middle| D_i = 1, Z_i \in \mathcal{Z}^{M_Z}\right\} \\ \vdots & \ddots & \vdots \\ \Pr\left\{X_i \in \mathcal{X}^{M_X} \middle| D_i = 0, Z_i \in \mathcal{Z}^1\right\} & \cdots & \Pr\left\{X_i \in \mathcal{X}^{M_X} \middle| D_i = 1, Z_i \in \mathcal{Z}^{M_Z}\right\} \end{pmatrix}$$

We can apply the Kleibergen-Paap rank test or the eigenvalue ratio estimator.

Kleibergen and Paap (2006); Ahn and Horenstein (2013)

back

Deriviation of feasible moment from Proposition 1

By substituting for $f_{Y(d)|U}$,

$$
\begin{aligned}
0 &= \mathbf{E}\left[m\big(Y_i(1), Y_i(0);\theta\big)\right] \\
&= \int_{\mathbb{R}}\int_{\mathbb{R}} m\left(y', y;\theta\right) f_{Y(0),Y(1)}\left(y, y'\right) dydy' \\
&= \int_{\mathbb{R}}\int_{\mathbb{R}} m\left(y', y;\theta\right) \left(\sum_{k=1}^{K} p_U(k) \cdot f_{Y(0)|U}\left(y|u\right) \cdot f_{Y(1)|U}\left(y'|u\right)\right) dydy' \\
&= \int_{\mathbb{R}}\int_{\mathbb{R}} m\left(y', y;\theta\right) \left(\sum_{k=1}^{K} p_U(k) \cdot \sum_{j=1}^{K} \tilde{\lambda}_{jk,0} f_{Y|D=0,Z}\left(y|z^{j}\right) \cdot \sum_{j'=1}^{K} \tilde{\lambda}_{j'k,1} f_{Y|D=1,Z}\left(y|z^{j'}\right)\right) dydy' \\
&= \sum_{k=1}^{K}\sum_{j=1}^{K}\sum_{j'=1}^{K} p_U(k)\tilde{\lambda}_{jk,0}\tilde{\lambda}_{j'k,1} \underbrace{\int_{\mathbb{R}}\int_{\mathbb{R}} m\left(y', y;\theta\right) \left(f_{Y|D=0,Z}\left(y|z^{j}\right) \cdot f_{Y|D=1,Z}\left(y|z^{j'}\right)\right) dydy'}_{\text{quadratic moment of } (Y_i, D_i, X_i, Z_i)}
\end{aligned}
$$

where $p_U(k) = \Pr\left\{U_i = u^k\right\}$

Deriviation of feasible moment from Proposition 1

For example,

$$
\begin{aligned}
F_{Y(1)-Y(0)}(\delta) &= \mathbf{E}\left[\mathbf{1}\{Y_i(1) \le Y_i(0) + \delta\}\right] \\
&= \sum_{k=1}^{K}\sum_{j=1}^{K}\sum_{j'=1}^{K} \frac{p_U(k)\tilde{\lambda}_{jk,1}\tilde{\lambda}_{j'k,0}}{p_{D,Z}(1,j)p_{D,Z}(0,j')} \\
&\qquad \cdot \mathbf{E}\left[\mathbf{1}\{Y_i \le Y_{i'} + \delta, D_i = 1, Z_i = z^j, D_{i'} = 0, Z_{i'} = z^{j'}\}\right]
\end{aligned}
$$

for all $\delta \in \mathbb{R}$, with $(Y_i, D_i, Z_i) \perp\!\!\!\perp (Y_{i'}, D_{i'}, Z_{i'})$.

The nuisance parameters are $\tilde{\lambda} = \text{vec}\left((\Lambda_0)^{-1}, (\Lambda_1)^{-1}\right)$ and

$$
p = \text{vec}\Big(\{p_{D,U}(d,k)\}_{d,k}, \{p_{D,Z}(d,j)\}_{d,j}\Big).
$$

$\tilde{\lambda}$ is matrix inverses of (Λ_0, Λ_1).

p collects joint probabilities of (D_i, U_i) and (D_i, Z_i); also a function of (Λ_0, Λ_1).

Deriviation of feasible moment from Proposition 1

$$
\begin{aligned}
& F_{Y(1)-Y(0)}(\delta) \\
&= \mathbf{E}\left[\Pr\left\{Y_i(1) \leq Y_i(0) + \delta | U_i\right\}\right] \\
&= \sum_{k=1}^{K} p_U(k) \underbrace{\int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}\{y \leq y' + \delta\} f_{Y(1)|U}(y|u^k) f_{Y(0)|U}(y'|u^k) dy dy'}_{=\Pr\left\{Y_i(1) \leq Y_i(0) + \delta | U_i = u^k\right\}} \quad \because Y_i(1) \perp\!\!\!\perp Y_i(0) \mid U_i \\
&= \sum_{k=1}^{K} p_U(k) \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}\{y \leq y' + \delta\} \left(\sum_{j=1}^{K} \tilde{\lambda}_{jk,1} f_{Y|D=1,Z}(y|z^j) \right) \quad \because \text{multiplying } \tilde{\Lambda}_d \text{ to } \mathbf{H}_d = \Gamma_d \cdot \Lambda_d \\
&\qquad \left(\sum_{j'=1}^{K} \tilde{\lambda}_{j'k,0} f_{Y|D=0,Z}(y'|z^{j'}) \right) dy dy' \\
&= \sum_{k=1}^{K} \sum_{j=1}^{K} \sum_{j'=1}^{K} p_U(u^k) \tilde{\lambda}_{jk,1} \tilde{\lambda}_{j'k,0} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}\{y \leq y' + \delta\} f_{Y|D=1,Z}(y|z^j) f_{Y|D=0,Z}(y'|z^{j'}) dy dy'
\end{aligned}
$$

back

Neyman orthogonality

Nuisance parameter estimation has first-order impact on DTE estimation.
Thus, I orthogonalize the feasible moment function $\tilde{m}$. Delta method

$\hat{\lambda}$ and $\hat{p}$ are highly nonlinear functions of $\mathbb{H}_0$ and $\mathbb{H}_1$.
No usual “first-order condition”-type moments available.

1. For $\tilde{\lambda}$, use the quadratic moments of conditional independence: for $d = 0, 1$,

$$\Pr\{Y_i(d) = y, X_i = x | U_i = u\} = \Pr\{Y_i(d) = y | U_i = u\} \cdot \Pr\{X_i = x | U_i = u\}.$$

more

2. For $p_{D,U}$, use the moments of law of iterated expectation:

$$\Pr\{D_i = d, X_i = x\} = \sum_{k=1}^{k} p_{D,U}(d,k) \Pr\left\{X_i = x \middle| U_i = u^k\right\}.$$

Neyman orthogonality

Let ϕ be the moment function for the additional moments.

Lemma 1. Assumptions 1-3 hold. Then, $\begin{pmatrix} \mathbf{E}\left[\frac{\partial}{\partial \tilde{\lambda}} \phi\right] \\ \mathbf{E}\left[\frac{\partial}{\partial p} \phi\right] \end{pmatrix}$ has full row rank.

With $\mu = \begin{pmatrix} \mathbf{E}\left[\frac{\partial}{\partial \tilde{\lambda}} \phi\right] \\ \mathbf{E}\left[\frac{\partial}{\partial p} \phi\right] \end{pmatrix}^{+} \begin{pmatrix} \mathbf{E}\left[\frac{\partial}{\partial \tilde{\lambda}} \tilde{m}\right] \\ \mathbf{E}\left[\frac{\partial}{\partial p} \tilde{m}\right] \end{pmatrix}$, the **orthogonalized moment function** is

$$
\begin{aligned}
&\psi\Big((Y_i, D_i, X_i, Z_i), (Y_{i'}, D_{i'}, X_{i'}, Z_{i'}); \theta, \tilde{\lambda}, p, \mu\Big) \\
&= \tilde{m}\Big((Y_i, D_i, X_i, Z_i), (Y_{i'}, D_{i'}, X_{i'}, Z_{i'}); \theta, \tilde{\lambda}, p\Big) - \mu^\intercal \phi\Big((Y_i, D_i, X_i, Z_i), (Y_{i'}, D_{i'}, X_{i'}, Z_{i'}); \tilde{\lambda}, p\Big)
\end{aligned}
$$

This applies to any moment-identified parameter in a finite mixture model. back

Principal component analysis vs. nonnegative matrix factorization

Principal component analysis:

- given a $M \times K$ matrix $\mathbf{H}$ and an integer $R > 0$, find a rank R matrix $\tilde{\mathbf{H}}$ such that

$$\min \left\| \mathbf{H} - \tilde{\mathbf{H}} \right\|_F$$

Nonnegative matrix factorization:

- given a $M \times K$ matrix $\mathbf{H}$ and an integer $R > 0$, find rank R ***nonnegative*** matrices Γ, Λ such that

$$\min \left\| \mathbf{H} - \Gamma \cdot \Lambda \right\|_F$$

NMF adds one more constraint: the low-rank representation should factor into nonnegative matrices.

back

Nonnegative matrix factorization

Γ_d can be further decomposed into Γ_X and $\Gamma_{Y(d)}$, using $Y_i(d) \perp\!\!\!\perp X_i \,\big|\, U_i$.

The minimization problem

$$\min \left\|\mathbb{H}_d - \Gamma_d \cdot \Lambda_d\right\|_F$$

becomes a quadratic program with linear constraints,
once we fix two out of the three matrices $\Gamma_X, \Gamma_{Y(d)}, \Lambda_d$.

Thus, find the (local) minima by iterating across three objects:

1. Given $\left(\Gamma_0^{(s)}, \Gamma_1^{(s)}\right)$, update (Λ_0, Λ_1).
2. Given $\left(\Lambda_0^{(s+1)}, \Lambda_1^{(s+1)}, {\Gamma_{Y(0)}}^{(s)}, {\Gamma_{Y(1)}}^{(s)}\right)$, update Γ_X.
3. Given $\left(\Lambda_0^{(s+1)}, \Lambda_1^{(s+1)}, {\Gamma_X}^{(s+1)}\right)$, update $\left(\Gamma_{Y(0)}, \Gamma_{Y(1)}\right)$.
4. Iterate **1**-**3** until convergence.

In practice, use may initial values to find the global minimum. back

Implementation: comparison to diagonalization

1. For each d, y, construct
$$\mathbb{H}_d(y)\Big(\sum_{y'} \mathbb{H}_d(y')\Big)^{-1}$$
where $\mathbb{H}_d(y)$ estimates $\Pr\{Y_i = y, X_i = x | D_i = d, Z_i = z\}$ and
$\sum_{y'} \mathbb{H}_d(y')$ estimates $\Pr\{X_i = x | D_i = d, Z_i = z\}$.
2. Diagonalize $\mathbb{H}_d(y)\Big(\sum_{y'} \mathbb{H}_d(y')\Big)^{-1}$ across d, y since
$$\mathbf{H}_d(y)\left(\sum_{y'} \mathbf{H}_d(y')\right)^{-1} = \Gamma_X \cdot \Delta_d(y) \cdot \Big(\Gamma_X\Big)^{-1}.$$
Sum-to-one will pin down eigenvectors, i.e. Γ_X.

back

Asymptotic normality + Delta method

The first-step NMF can be thought of as a (loosely defined) GMM or MLE estimator.

1. Nonnegativity constraints often bind.

   The estimators may not be asymptotically normal.
   e.g. Bonhomme et al. (2016) derives asymptotic normality, while not imposing nonnegativity.

2. DTE parameters are highly nonlinear in Λ_0 and Λ_1.

   Asymptotic normality using Delta method may converge slowly.
   Bootstrapping for standard error $\Rightarrow$ higher computation burden, given NMF.

3. Orthogonalization may help with discretization bias when U_i is continuous.

   We would need $\widehat{\Lambda}_d$ and $\left(\widehat{\Lambda}_d\right)^{-1}$ to converge to true bivariate functions at $n^{-\frac{1}{4}}$ rate.

Hence, $\sqrt{n}$-consistency and orthogonalization. back

Additional moments in orthogonalization

$\Pr\{Y_i = y, X_i = x | U_i = u\} = \Pr\{Y_i = y | U_i = u\} \cdot \Pr\{X_i = x | U_i = u\}$ implies

$$
\begin{aligned}
&\frac{1}{2}\sum_{l=1}^{K}\frac{\tilde{\lambda}_{lk,d}}{p_{D,Z}(d,l)} \cdot \mathbf{E}\left[\mathbf{1}\{Y_i = y, D_i = d, X_i = x, Z_i = z^l\}\right] \\
&+ \frac{1}{2}\sum_{m=1}^{K}\frac{\tilde{\lambda}_{mk,d}}{p_{D,Z}(d,m)} \cdot \mathbf{E}\left[\mathbf{1}\{Y_j = y, D_j = d, X_j = x, Z_i = z^m\}\right] \\
&- \frac{1}{2}\sum_{l=1}^{K}\sum_{m=1}^{K}\frac{\tilde{\lambda}_{lk,d}\tilde{\lambda}_{mk,d}}{p_{D,Z}(d,l) \cdot p_{D,Z}(d,m)}\mathbf{E}\left[\mathbf{1}\{Y_i = y, D_i = d, Z_i = z^l, X_j = x, D_j = d, Z_j = z^m\}\right] \\
&- \frac{1}{2}\sum_{l=1}^{K}\sum_{m=1}^{K}\frac{\tilde{\lambda}_{lk,d}\tilde{\lambda}_{mk,d}}{p_{D,Z}(d,l) \cdot p_{D,Z}(d,m)}\mathbf{E}\left[\mathbf{1}\{X_i = x, D_i = d, Z_i = z^l, Y_j = y, D_j = d, Z_j = z^m\}\right] = 0
\end{aligned}
$$

with $(Y_i, D_i, Z_i) \perp\!\!\!\perp (Y_j, D_j, Z_j)$. back

Regularity conditions for infeasible GMM model

$\theta \in \Theta \subset \mathbb{R}^p$ and $m \in \mathbb{R}^L$. Some weighting matrix $A \in \mathbb{R}^{L \times L}$ is given.

Let W_i^* denote $(Y_i(1), Y_i(0), D_i, X_i)$ and assume

i. Θ is compact and the true parameter θ^0 in the interior of Θ.

ii. θ^0 uniquely solves $\mathbf{E}\left[m(W_i^*; \theta)\right] = \mathbf{0}$.

iii. $m_l(W_i^*; \theta)$ is (almost surely) continuous in $\theta \in \Theta$ for $l = 1, \cdots, L$.

iv. $\mathbf{E}\left[\left\| \frac{\partial}{\partial \theta} m_l(W_i^*; \theta)\big|_{\theta=\theta^0} \right\|^2\right] < \infty$ for $l = 1, \cdots, L$.

v. $\frac{\partial^2}{\partial \theta \partial \theta^\intercal} \mathbf{E}\left[m(W_i^*; \theta)^\intercal A m(W_i^*; \theta)\right]$ is continuous in θ and $\frac{\partial^2}{\partial \theta \partial \theta^\intercal} \mathbf{E}\left[m(W_i^*; \theta)^\intercal A m(W_i^*; \theta)\right]\big|_{\theta=\theta^0}$ is positive definite.

vi. $\exists$ a function G s.t. $\sup_l |m_l(w; \theta)| \le G(w)$ for all $\theta \in \Theta$ and $\mathbf{E}\left[G(W_i^*)\right] < \infty$.

vii. $\exists$ a function B s.t. for any θ^1, θ^2,
$\left\| \frac{\partial}{\partial \theta} \left(m(w; \theta)^\intercal A m(w; \theta)\right) \big|_{\theta=\theta^1} - \frac{\partial}{\partial \theta} \left(m(w; \theta)^\intercal A m(w; \theta)\right) \big|_{\theta=\theta^2} \right\| \le B(w) \|\theta^1 - \theta^2\|$
and $\mathbf{E}\left[B(W_i^*)^2\right] < \infty$.

back

Falsification test

$Y_i(1) \perp\!\!\!\perp Y_i(0) \mid U_i$ from Assumption 2 is fundamentally untestable.

Instead, I test $X_i \perp\!\!\!\perp D_i \mid U_i$ with estimators assuming $Y_i(d) \perp\!\!\!\perp X_i \mid U_i$.

"Can we construct a latent variable U_i that satisfies 1) conditional independence $Y_i(d) \perp\!\!\!\perp X_i \mid U_i$ and 2) random treatment $X_i \perp\!\!\!\perp D_i \mid U_i$?"

For this test, do not impose $X_i \perp\!\!\!\perp D_i \mid U_i$ in the NMF.

In the short panel context,

- cannot test the conditional independence *across treatment regime*.
- can somewhat test the *intertemporal* conditional independence, given random treatment.

back

Data generating process

The specifics of the DGPs are as follows:

- $n = 500, 1000, 2000$.
- $D_i \perp\!\!\!\perp \big(Y_i(1), Y_i(0), X_i, Z_i, U_i\big)$ and $\Pr\{D_i = 1\} = 0.5$.
- $(p_U(1), p_U(2), p_U(3)) = (0.3, 0.4, 0.3)$.
- Since Y_i is continuous, a three-way partition using 1/3-th and 2/3-th quantile was used.
- The conditional probabilities of the discretized Y_i are

$$\Gamma_{Y(0)} = \begin{pmatrix} 0.901 & 0.612 & 0.237 \\ 0.098 & 0.359 & 0.579 \\ 0.002 & 0.029 & 0.184 \end{pmatrix} \quad \text{and} \quad \Gamma_{Y(1)} = \begin{pmatrix} 0.209 & 0.043 & 0.000 \\ 0.396 & 0.417 & 0.115 \\ 0.395 & 0.540 & 0.885 \end{pmatrix}$$

back

Data generating process

Conditional distribution of $Y_i(1) - Y_i(0)$ given U_i:

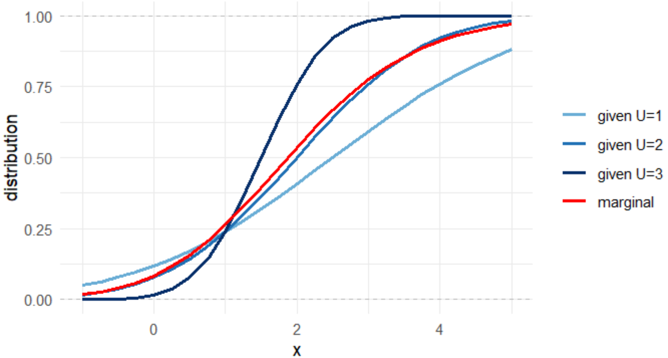

Figure 4: Marginal and conditional distributions of $Y_i(1) - Y_i(0)$.

back

Choice of K

1. Consider a $M_X \times 2M_Z$ matrix $\mathbf{H}_X$:

$$\mathbf{H}_X = \begin{pmatrix} \Pr\left\{X_i \in \mathcal{X}^1 | D_i = 0, Z_i \in \mathcal{Z}^1\right\} & \cdots & \Pr\left\{X_i \in \mathcal{X}^1 | D_i = 1, Z_i \in \mathcal{Z}^{M_Z}\right\} \\ \vdots & \ddots & \vdots \\ \Pr\left\{X_i \in \mathcal{X}^{M_X} | D_i = 0, Z_i \in \mathcal{Z}^1\right\} & \cdots & \Pr\left\{X_i \in \mathcal{X}^{M_X} | D_i = 1, Z_i \in \mathcal{Z}^{M_Z}\right\} \end{pmatrix}.$$

   $\mathbf{H}_X$ pools information from $\{i : D_i = 0\}$ and $\{i : D_i = 1\}$ and should be at most rank K.
   Apply eigenvalue ratio estimator and rank test.

2. With true K, estimated densities should satisfy

$$f_{X|D=1,U}(x|u) = f_{X|D=0,U}(x,u) \qquad \forall x, u.$$

   Apply falsification tests.

back

Choice of K: rank test

Both rank test and eigenvalue ratio estimator suggest $K = 3$.

K	1	2	3	4	5	6	7	8
eigenvalue ratio	3.505	3.991	4.029	2.721	1.653	1.863	1.418	3.309
growth ratio	0.964	1.135	1.472	1.353	0.893	0.956	0.580	1.035

Table 4: Eigenvalue ratios and growth ratios

K	1	2	3	4	5	6
test statistic	884.82	116.23	35.75	20.08	13.80	7.94
p-value	0.000	0.001	0.984	0.998	0.995	0.992

Table 5: Kleibergen-Paap rank test statistics for H_0 : rank $= K$ and their p-values

back

Choice of K: falsification test

Falsification test statistic T_n for $f_{X|D=1,U}(x|u) = f_{X|D=0,U}(x,u) \quad \forall x, u$.

K	3	4	5	6
T_n	17.68	27.07	16.79	47.66
p-value	0.477	0.301	0.975	0.092

Table 6: Falsification test statistic T_n and their p-values

back

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